

MOHAMMED SAHAL SHAIKH

Graduate Mechanical Engineer

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South African Permanent Resident | ECSA Candidate Engineer registration in progress | Code 10 driver's licence (in progress)

PROFESSIONAL SUMMARY

Mechanical Engineering (Honours) graduate (Curtin University, 2026) with five peer-reviewed publications (Springer Nature, IEEE, MARLOG) and a CFD-validated Honours thesis. Combines design (SolidWorks, Inventor), simulation (ANSYS Fluent/Mechanical, MATLAB) and hands-on fabrication, from building an instrumented solar test rig to airframe maintenance training on the Piper PA-28. Founded four university engineering chapters and led international programmes spanning four Curtin campuses and 12 UAE universities. Seeking a graduate mechanical, simulation, R&D or energy engineering role in Gauteng; incoming MSc (University of Pretoria, 2027).

EDUCATION

BEng (Honours), Mechanical Engineering | Curtin University, Dubai 2022 – 2026

- Upper Second Class Honours • **CWA 72.89%** • **Final-year average 78%** • FEA and CFD units both above 80%
- Coursework: Computational Fluid Dynamics, Finite Element Analysis, Thermodynamics, Heat Transfer, Engineering Management

MSc Mechanical Engineering (incoming) | University of Pretoria, CERG-FLUX Lab 2027

- Proposed research: pore-scale Lattice Boltzmann simulation of bio-inspired porous structures for phase-change thermal energy storage (supervisor: Prof. M. Bhamjee).

TECHNICAL SKILLS

CAD & Design: SolidWorks, Autodesk Inventor Professional, AutoCAD, Onshape, FreeCAD; engineering drawings, GD&T (ASME)

Simulation & Analysis: ANSYS Fluent, ANSYS Mechanical, MATLAB, Simulink, SimScale; CFD, FEA, heat transfer, fatigue & reliability analysis

Programming & Data: Python (OpenCV, scikit-learn), Arduino IDE, Advanced Excel, LaTeX; openLCA (LCA), DWSIM (process modelling)

Manufacturing & Test: Test-rig design and build, instrumentation & data acquisition, sheet metal, 3D printing, laser cutting, soldering

Electronics (EDA): LTspice, Cadence, Tinkercad

Languages: English, Afrikaans, Urdu (fluent); Turkish (conversational)

PROFESSIONAL EXPERIENCE

Solar Energy Research Intern | Transnational Academic Group & Curtin University Feb 2026 – Jun 2026

- Designed (SolidWorks), built and instrumented a solar-panel soiling test rig, translating a research question into a controlled experiment quantifying dust-driven losses in efficiency and service life.
- Ran controlled tests and supporting CFD analysis; the dataset informed panel cleaning and maintenance scheduling.
- Synthesised **20+ peer-reviewed papers** into a publication-standard technical report on degradation mechanisms.

STEM / CAD Instructor | The Logix Engine Oct 2025 – Feb 2026

- Coached a FIRST LEGO League team from concept to completion across the robot and innovation challenges.
- Taught CAD (Onshape, Inventor, AutoCAD), robotics and prototyping at multiple schools; authored new project-based CAD and robotics courses for senior students.

Project Coordinator, ICISAS 2025 | Transnational Academic Group Dec 2024 – Mar 2025

- Screened and triaged **450+ research submissions** against scope and quality criteria for an international engineering conference.
- Coordinated the peer-review workflow across **1,000+ reviewers**, managing reviewer assignment and the end-to-end submission system.

Materials & Solar Research Intern | Transnational Academic Group & Curtin University Aug 2023 – Mar 2024

- Benchmarked **5 solar-panel technologies** and evaluated **2 aluminium-alloy combinations**, assessing how manufacturing route affects mechanical properties.

Aviation Maintenance Intern | Sarsan Aviation Academy (GCAA CAR-147), Dubai Aug 2023

- Hands-on airframe maintenance on the Piper PA-28 across **26 aircraft components**: sheet-metal repair, safety wiring, wing attachment and flight-control rigging; trained in gas turbines, avionics and aviation regulations.

SELECTED PROJECTS

Bio-Inspired Oil Sorbent Characterisation | Honours Thesis

2025 – 2026

- Built a Python/OpenCV image-analysis pipeline across 59 cross-sections, detecting **4,498 pores** and deriving porosity (22.2%), mean pore diameter (954 μm) and permeability ($9.16 \times 10^{-11} \text{ m}^2$).
- Built a 3D porous-zone ANSYS Fluent model (Darcy–Forchheimer) predicting permeability **within 0.17%** of the image-derived value; gravimetric testing on 26 specimens reached **96.6%** of theoretical maximum uptake.

Aviation Decarbonisation Diagnostic ML Framework | Lead author, ICISAS 2026 (IEEE)

2026

- Ran Boeing Cascade scenario ensembles (SAF, hydrogen, electric, combined) with an ML surrogate and Sobol sensitivity analysis; identified configuration faults and turned them into a diagnostic validation framework. Presented at DICEGAI 2026.

Thassos Marble–Aluminium Passive Cooling • CNT-Enhanced SAF (TEA/LCA) | Lead author, ICISAS 2026

2026

- Supervise junior researchers on two further papers: ANSYS thermal/mechanical analysis of a marble–aluminium composite for data-centre cooling (diagnosed five simulation faults, defined three optimisation pathways), and an openLCA/DWSIM lifecycle and techno-economic study of CNT-enhanced synthetic SAF.

Industrial Gearbox Design | Curtin University

2024

- Designed a 1.6 kW gearbox for an 8-year life at <1% failure probability: stress, fatigue and reliability analysis with full SolidWorks assembly and drawing package.

Solar Cooking Backpack | James Dyson Award, National Finalist (team lead)

2023

- Led design of a portable solar cooker for off-grid communities; prototype cooked a full meal in ~2 hours, ~65% faster than benchmark solar cookers.

LEADERSHIP

- **Founded four student chapters** at Curtin University Dubai: IEEE, ASHRAE, Robotics Club and FEDA.
- **President, IEEE Student Branch** (Jan 2025 – Jun 2026): organised IEEE Tech Fest (AI, robotics, ethical hacking) and IEEE Convergence with Curtin Colombo, including its first Simulation Engineering event; coordinated the inaugural IEEE Curtin ASPIRE 2026 SDG Innovation Challenge across **4 campuses** (Dubai, Colombo, Malaysia, Bentley).
- **President, ASHRAE Student Branch** (Jul 2025 – Jun 2026): ran the ASHRAE Poster Competition for three years; the 2025 edition drew **40+ entries from 12 UAE universities**.
- **President, PMI UAE Student Club** (Aug 2025 – Dec 2026): led a governance restructure with two bylaw amendments; organised a career-readiness event for **50–60 participants**, owning budget, speakers and logistics.
- **Founding President, Curtin Robotics Club** (led team to 1st place, Mubarmij50) • **Treasurer, Engineers Australia** (3-year budget oversight) • **Events Officer, ASME** • **Engineering School Officer**, Student Council.

PUBLICATIONS

- Gadher, S., **Shaikh, M.S.M.S.**, Selvan, C.P. (2025). Effects on the Microstructure and Mechanical Properties of the AISi14 Alloy Produced by the Selective Laser Melting (SLM) Technique. ICISAS 2023, Springer Nature. doi:10.1007/978-3-031-68952-9_15
- Selvan, C.P., Nayar, A., Nigaraja, M.R., **Sahal, M.** (2026). Integrating Climate Resilience and Circular Economy Principles in Developing Green-Blue Logistic Corridors for a Zero-Emission Future. MARLOG 15, International Maritime Transport and Logistics Journal. doi:10.21622/MARLOG.2026.15.1.89
- Three co-authored IEEE ASET 2025 papers (interdisciplinary: ECG signal processing, medical-data diagnosis, cloud data security). doi:10.1109/ASET66891.2025.11427994 • .11428087 • .11427882

AWARDS, CERTIFICATIONS & TRAINING

- **1st Place**, Mubarmij50 Robotics Competition (2024) • **1st Place**, Manipal University Hydro Jet Competition (2022)
- **1st Runner-Up**, Western Australia Government Climate Action Challenge (2024) • **Winner**, ASHRAE Falcon Debate
- Ansys: Supersonic, Subsonic & 2D Flows; Shock-Expansion Theory; Flow over an Airfoil (2026) • ASME: GD&T Drawing Interpretation, Design for Sustainability, Industry 4.0
- RAeS & Vernewell 5-week aerospace mentorship • GCAA CAR-147 maintenance training • Microbotics, CTU Training Solutions (2019)

REFERENCE

Prof. Chithirai Pon Selvan

Head of School, Science and Engineering, Curtin University Dubai

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